

LA TUQUE, QC, Canada

WMO: 713780

Lat: 47.411N

Lon: 72.787W

Elev: 169

StdP: 99.31

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.2	-28.2	-34.8	0.1	-30.7	-32.0	0.2	-27.7	8.1	-10.2	7.3	-10.4	0.8	110	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	30.2	19.3	28.2	18.4	26.5	17.7	21.7	26.3	20.6	25.0	19.6	23.7	3.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.2	23.6	19.2	14.3	22.4	18.2	13.3	21.3	64.2	26.2	60.3	24.8	56.7	23.6	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.4	5.7	-35.9	33.7	2.7	1.8	-37.8	35.0	-39.4	36.1	-40.9	37.1	-42.9	38.4		
			-36.0	23.7	2.7	1.1	-38.0	24.4	-39.6	25.1	-41.2	25.7	-43.1	26.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-14.1	-12.3	-5.6	2.6	10.8	16.3	18.9	17.6	13.2	6.1	-1.2	-9.1	(6)
	DBStd	12.67	7.87	7.20	6.88	4.68	4.48	4.02	3.04	3.26	4.28	4.39	5.40	7.05	(7)
	HDD10.0	3204	748	623	485	226	43	2	0	15	136	336	591		(8)
	HDD18.3	5458	1006	857	743	473	236	87	30	52	162	378	585	849	(9)
	CDD10.0	903	0	0	0	3	69	190	276	237	112	16	1	0	(10)
	CDD18.3	114	0	0	0	0	3	24	47	31	9	0	0	0	(11)
	CDH23.3	1313	0	0	1	2	102	331	486	303	85	3	0	0	(12)
	CDH26.7	356	0	0	0	0	26	106	134	69	21	1	0	0	(13)
(14) Wind	WSAvg	2.3	2.3	2.3	2.7	2.7	2.5	2.3	2.1	2.0	2.1	2.3	2.4	2.2	(14)
(15) Precipitation	PrecAvg	959	57	58	60	61	76	91	119	104	97	75	84	87	(15)
	PrecMax	1238	96	128	150	116	251	168	294	153	167	149	198	144	(16)
	PrecMin	844	21	2	12	18	16	51	54	36	48	33	35	31	(17)
	PrecStd	106	19	38	29	24	48	31	54	32	30	30	33	31	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	5.6	12.6	21.0	29.8	32.4	32.6	31.2	29.3	22.1	14.3	6.1	(19)
		MCWB	5.8	3.1	6.3	11.6	15.9	19.8	21.1	19.6	19.8	14.9	11.9	5.0	(20)
	2%	DB	2.7	2.7	8.3	17.0	26.2	29.7	29.9	28.6	25.6	18.4	10.9	3.5	(21)
		MCWB	1.9	0.6	3.6	8.7	14.5	18.7	20.0	18.9	18.5	13.3	8.6	2.5	(22)
	5%	DB	0.6	0.9	5.8	13.8	23.3	27.2	27.9	26.7	23.1	15.5	8.5	1.7	(23)
		MCWB	-0.3	-0.4	2.1	7.0	13.2	17.3	18.6	18.3	17.0	12.1	6.5	0.8	(24)
	10%	DB	-2.2	-0.7	3.6	10.8	20.3	24.7	26.1	24.7	20.9	13.0	6.1	0.5	(25)
		MCWB	-3.1	-2.0	1.0	5.6	12.2	16.0	18.0	17.5	16.2	10.1	4.4	-0.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	4.1	8.1	13.7	19.3	22.6	23.4	22.2	21.7	16.7	12.7	5.2	(27)
		MCDB	6.4	5.1	10.4	18.8	24.1	28.7	28.8	26.8	26.5	20.2	14.3	5.8	(28)
	2%	WB	2.1	1.2	4.2	10.0	17.1	20.8	21.9	21.0	19.9	14.5	9.2	2.7	(29)
		MCDB	2.6	2.2	6.8	13.8	22.4	25.7	26.3	25.4	23.6	17.4	10.5	3.4	(30)
	5%	WB	-0.3	-0.2	2.7	8.0	15.3	19.3	20.7	20.0	18.4	12.6	6.7	1.1	(31)
		MCDB	0.6	0.9	5.2	11.9	19.8	23.9	25.1	24.2	21.9	15.2	8.2	1.6	(32)
	10%	WB	-3.0	-2.1	1.3	6.2	13.5	18.0	19.6	18.9	16.9	10.8	4.7	-0.3	(33)
		MCDB	-2.2	-0.6	3.3	10.1	18.7	22.5	23.8	22.5	20.1	12.7	6.1	0.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	13.1	12.4	11.7	13.9	13.3	12.3	12.2	11.6	9.3	7.7	8.9	(35)
		MCDBR	10.2	10.0	13.7	18.4	19.7	17.5	15.4	15.0	14.4	13.5	10.9	7.3	(36)
	5% WB	MCWBR	9.9	8.7	9.4	10.6	9.2	7.6	6.8	6.7	7.8	8.5	8.9	6.9	(37)
		MCWBR	10.0	9.3	11.6	14.8	14.4	12.8	12.6	11.7	12.4	11.6	9.9	6.9	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.270	0.278	0.309	0.341	0.361	0.388	0.396	0.386	0.365	0.336	0.302	0.263	(40)
		taud	2.268	2.245	2.230	2.378	2.422	2.351	2.334	2.371	2.423	2.495	2.486	2.360	(41)
	Edn at Noon	Edn at Noon	811	888	909	913	903	876	862	857	843	818	780	782	(42)
		Edh at Noon	76	97	116	112	111	120	121	112	97	77	63	62	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.35	2.34	3.66	4.61	5.10	5.49	5.28	4.59	3.55	2.05	1.17	0.92	(44)
	RadStd	RadStd	0.22	0.35	0.35	0.39	0.51	0.43	0.53	0.24	0.28	0.19	0.16	0.07	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page